

Assignment 10: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1
#load needed packages
library(tidyverse)
library(here);
here() #pointing me to the Assignments folder where I am currently working
```

```
## [1] "/home/guest/ENV872/EDE_Spring2026/Assignments"
```

```
library(rvest)
library(lubridate)
library(dplyr)
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):
 - Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
 - Set the year to 2024
 - Scroll down and select the LWSP link next to Durham Municipality.

- Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```
#2
#indicate the NC Water site URL to be scraped
webpage <- read_html('https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024')
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
 - Water system name
 - Contact Person
 - Ownership
- From the “3. Water Supply Sources” section:
 - Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “Reginald Hicks”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

```
#3
#scrape the water system name
water_system_name <- webpage %>%
  html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>%
  html_text()
water_system_name
```

```
## [1] "Durham"
```

```
#scrape the contact person
contact_person <- webpage %>%
  html_nodes("div+ table tr:nth-child(4) td:nth-child(2)") %>%
  html_text()
contact_person
```

```
## [1] "Reginald Hicks"
```

```
#scrape the ownership
ownership <- webpage %>%
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>%
  html_text()
ownership
```

```
## [1] "Municipality"
```

```
#scrape maximum day use for each month
```

```
MGD_month <- webpage %>%
  html_nodes("th~ td+ td") %>%
  html_text()
MGD_month
```

```
## [1] "34.5000" "36.0600" "37.3300" "32.1000" "46.6500" "37.3600" "38.2000"
## [8] "41.9000" "36.5800" "36.7300" "42.9600" "34.4500"
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It's likely you won't be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc. . .

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

```
#4
#convert scraped data into a dataframe
df_withdrawals <- data.frame("Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),
                             "WaterSystemName" = rep(water_system_name),
                             "ContactPerson" = rep(contact_person),
                             "Ownership" = rep(ownership),
                             "MaximumDailyWithdrawl" = as.numeric(MGD_month),
                             "Year" = rep(2024))

#add date column with month and year in data format
#I am not aure what data format means. this comes from instructions.
#I will assume Date format

df_withdrawals <- df_withdrawals %>%
  mutate(Date = as.Date(paste(Year, Month, "01", sep = "-")))
class(df_withdrawals$Date) #make sure it is still in date format
```

```
## [1] "Date"
```

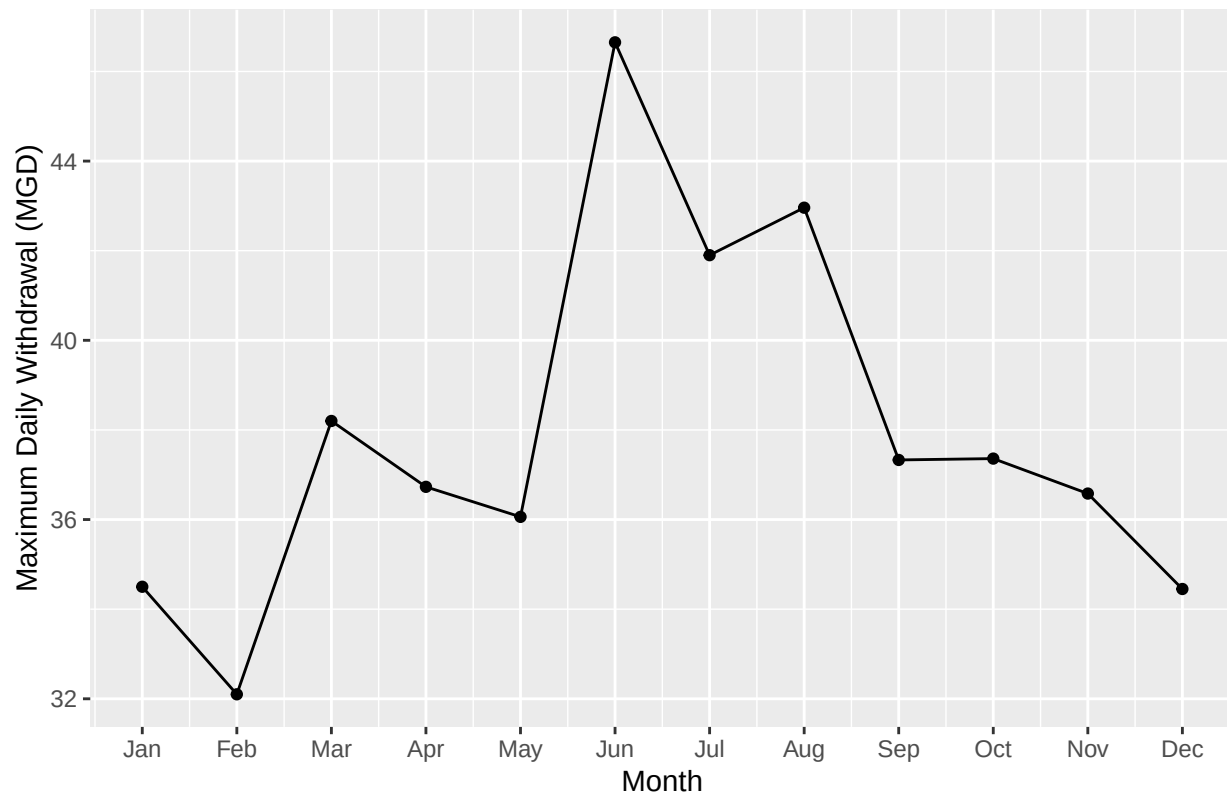
```
show(df_withdrawals)
```

```
##      Month WaterSystemName ContactPerson Ownership MaximumDailyWithdrawl Year
## 1      1      Durham Reginald Hicks Municipality      34.50 2024
## 2      5      Durham Reginald Hicks Municipality      36.06 2024
## 3      9      Durham Reginald Hicks Municipality      37.33 2024
## 4      2      Durham Reginald Hicks Municipality      32.10 2024
## 5      6      Durham Reginald Hicks Municipality      46.65 2024
## 6     10      Durham Reginald Hicks Municipality      37.36 2024
```

```
## 7      3      Durham Reginald Hicks Municipality      38.20 2024
## 8      7      Durham Reginald Hicks Municipality      41.90 2024
## 9     11      Durham Reginald Hicks Municipality      36.58 2024
## 10     4      Durham Reginald Hicks Municipality      36.73 2024
## 11     8      Durham Reginald Hicks Municipality      42.96 2024
## 12    12      Durham Reginald Hicks Municipality      34.45 2024
##           Date
## 1  2024-01-01
## 2  2024-05-01
## 3  2024-09-01
## 4  2024-02-01
## 5  2024-06-01
## 6  2024-10-01
## 7  2024-03-01
## 8  2024-07-01
## 9  2024-11-01
## 10 2024-04-01
## 11 2024-08-01
## 12 2024-12-01
```

```
#5
#create line plot of maximum daily withdrawals across months
df_withdrawals <-
  ggplot(df_withdrawals, aes(x = Month, y = MaximumDailyWithdrawl)) +
  geom_line() +
  geom_point() +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  #making sure months show on x axis since I got points off for that last hw
  labs(
    title = "Maximum Daily Withdrawals Across Months",
    x = "Month",
    y = "Maximum Daily Withdrawal (MGD)"
  )
show(df_withdrawals)
```

Maximum Daily Withdrawals Across Months



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs, “PWSID” and “year”, that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function’s output

```
#6.
#construct a scraping function
scrape.it <- function(the_year, the_PWSID){
  #Get the proper url
  the_url <-
    paste0('https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=', the_PWSID, '&year=', the_year)
  print(the_url)

  #Fetch the website
  the_website <- read_html(the_url)

  #Scrape the data
  #<enter code from above here>
  the_water_system_name <- the_website %>%
    html_nodes('div+ table tr:nth-child(1) td:nth-child(2)') %>% html_text()
  the_contact_person <- the_website %>%
```

```

    html_nodes('div+ table tr:nth-child(4) td:nth-child(2)') %>% html_text()
the_ownership <- the_website %>%
    html_nodes('div+ table tr:nth-child(2) td:nth-child(4)') %>% html_text()
the_MGD_month <- the_website %>% html_nodes('th~ td+ td') %>% html_text()

#Convert to dataframe
#<enter code from above here, adding a column for the month>
the_df <- data.frame(
  Month = c(1,5,9,2,6,10,3,7,11,4,8,12),
  Water_System_Name = the_water_system_name,
  Contact_Person = the_contact_person,
  Ownership = the_ownership,
  Maximum_Daily-Withdrawal = the_MGD_month,
  Year = rep(the_year))

#Return the dataframe
return(the_df)
}

# test the above function
df2024 <- scrape.it(2024, "03-32-010")

```

```
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024"
```

```
show(df2024)
```

```
##      Month Water_System_Name Contact_Person      Ownership Maximum_Daily-Withdrawal
## 1         1      Durham Reginald Hicks Municipality      34.5000
## 2         5      Durham Reginald Hicks Municipality      36.0600
## 3         9      Durham Reginald Hicks Municipality      37.3300
## 4         2      Durham Reginald Hicks Municipality      32.1000
## 5         6      Durham Reginald Hicks Municipality      46.6500
## 6        10      Durham Reginald Hicks Municipality      37.3600
## 7         3      Durham Reginald Hicks Municipality      38.2000
## 8         7      Durham Reginald Hicks Municipality      41.9000
## 9        11      Durham Reginald Hicks Municipality      36.5800
## 10        4      Durham Reginald Hicks Municipality      36.7300
## 11        8      Durham Reginald Hicks Municipality      42.9600
## 12       12      Durham Reginald Hicks Municipality      34.4500
##      Year
## 1    2024
## 2    2024
## 3    2024
## 4    2024
## 5    2024
## 6    2024
## 7    2024
## 8    2024
## 9    2024
## 10   2024
## 11   2024
## 12   2024
```

- Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```
#7
#use the function to pull max daily withdrawals in 2020
df2020 <- scrape.it(2020, "03-32-010")
```

```
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2020"
```

```
show(df2020)
```

```
##      Month Water_System_Name Contact_Person  Ownership Maximum_Daily_Withdrawal
## 1         1          Durham Reginald Hicks Municipality          36.0100
## 2         5          Durham Reginald Hicks Municipality          36.9800
## 3         9          Durham Reginald Hicks Municipality          41.6900
## 4         2          Durham Reginald Hicks Municipality          32.0500
## 5         6          Durham Reginald Hicks Municipality          40.6100
## 6        10          Durham Reginald Hicks Municipality          40.5600
## 7         3          Durham Reginald Hicks Municipality          37.2900
## 8         7          Durham Reginald Hicks Municipality          43.6300
## 9        11          Durham Reginald Hicks Municipality          33.3200
## 10        4          Durham Reginald Hicks Municipality          32.3700
## 11        8          Durham Reginald Hicks Municipality          41.9300
## 12       12          Durham Reginald Hicks Municipality          28.0600
##      Year
## 1    2020
## 2    2020
## 3    2020
## 4    2020
## 5    2020
## 6    2020
## 7    2020
## 8    2020
## 9    2020
## 10   2020
## 11   2020
## 12   2020
```

```
#show the structure of the dataframe
glimpse(df2020)
```

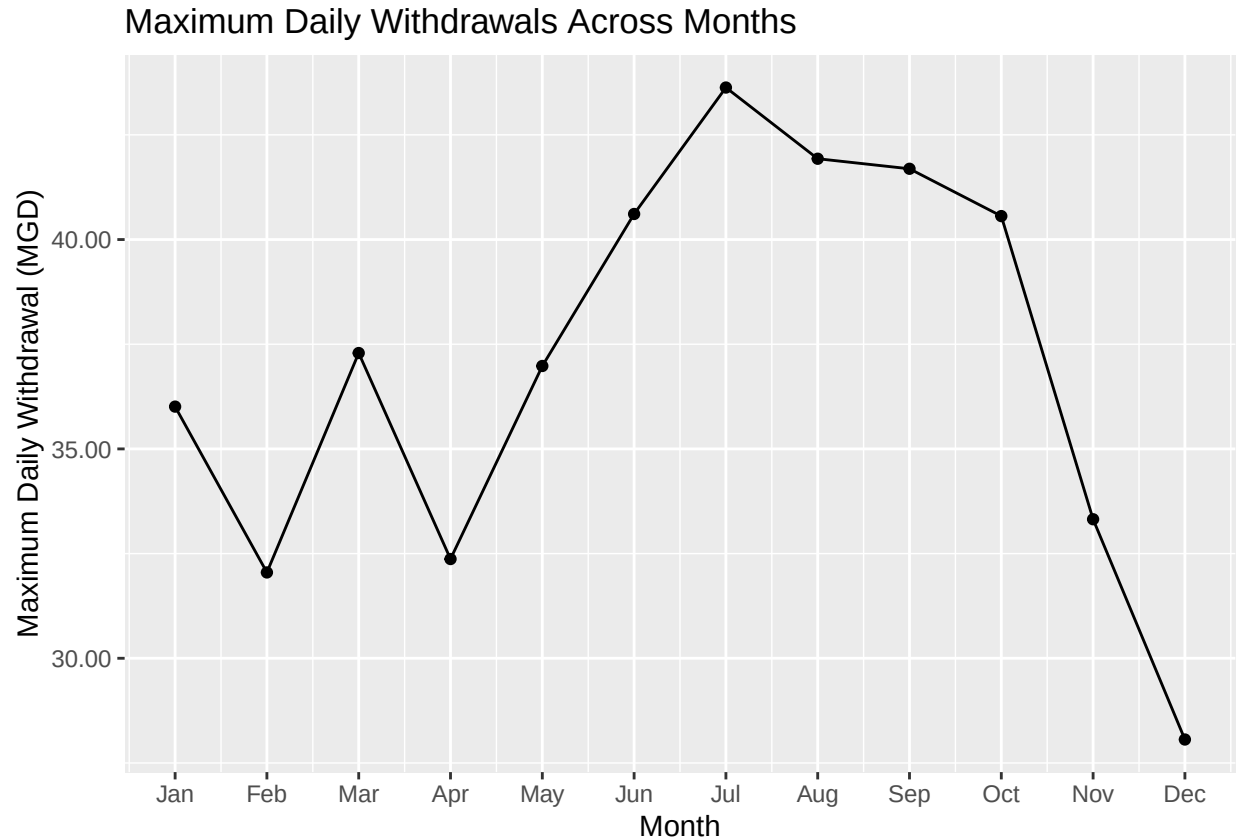
```
## Rows: 12
## Columns: 6
## $ Month          <dbl> 1, 5, 9, 2, 6, 10, 3, 7, 11, 4, 8, 12
## $ Water_System_Name <chr> "Durham", "Durham", "Durham", "Durham", "Durh~
## $ Contact_Person   <chr> "Reginald Hicks", "Reginald Hicks", "Reginald~
## $ Ownership        <chr> "Municipality", "Municipality", "Municipality~
## $ Maximum_Daily_Withdrawal <chr> "36.0100", "36.9800", "41.6900", "32.0500", "~
## $ Year            <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 202~
```

```

#make sure MGD_month is numeric to avoid errors down the line
df2020$Maximum_Daily-Withdrawal <- as.numeric(df2020$Maximum_Daily-Withdrawal)

#code to plot max daily withdrawals
df_withdrawals_2020 <-
  ggplot(df2020, aes(x = Month, y = Maximum_Daily-Withdrawal, group=1)) +
  geom_line() +
  geom_point() +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  scale_y_continuous(labels = scales::label_number(accuracy = 0.01)) +
  #making sure months show on x axis since I got points off for that last hw
  labs(
    title = "Maximum Daily Withdrawals Across Months",
    x = "Month",
    y = "Maximum Daily Withdrawal (MGD)"
  )
show(df_withdrawals_2020)

```



- Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```

#8
#use function to extract data for Cary
dfC2020 <- scrape.it(2020, "03-92-020")

```



```
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-92-020&year=2020"
```

```
show(dfC2020)
```

```
##      Month Water_System_Name Contact_Person  Ownership Maximum_Daily_Withdrawal
## 1         1          Cary    Jamie Revels Municipality          21.9500
## 2         5          Cary    Jamie Revels Municipality          23.2200
## 3         9          Cary    Jamie Revels Municipality          25.4900
## 4         2          Cary    Jamie Revels Municipality          17.7000
## 5         6          Cary    Jamie Revels Municipality          26.1900
## 6        10          Cary    Jamie Revels Municipality          23.6000
## 7         3          Cary    Jamie Revels Municipality          20.5600
## 8         7          Cary    Jamie Revels Municipality          28.0100
## 9        11          Cary    Jamie Revels Municipality          21.9000
## 10        4          Cary    Jamie Revels Municipality          19.3200
## 11        8          Cary    Jamie Revels Municipality          25.4000
## 12       12          Cary    Jamie Revels Municipality          19.0200
##      Year
## 1    2020
## 2    2020
## 3    2020
## 4    2020
## 5    2020
## 6    2020
## 7    2020
## 8    2020
## 9    2020
## 10   2020
## 11   2020
## 12   2020
```

```
#show the structure of the dataframe
```

```
glimpse(dfC2020)
```

```
## Rows: 12
## Columns: 6
## $ Month          <dbl> 1, 5, 9, 2, 6, 10, 3, 7, 11, 4, 8, 12
## $ Water_System_Name <chr> "Cary", "Cary", "Cary", "Cary", "Cary", "Cary~
## $ Contact_Person   <chr> "Jamie Revels", "Jamie Revels", "Jamie Revels~
## $ Ownership        <chr> "Municipality", "Municipality", "Municipality~
## $ Maximum_Daily_Withdrawal <chr> "21.9500", "23.2200", "25.4900", "17.7000", "~
## $ Year            <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 202~
```

```
#change MGD column to numeric because it caused problems
```

```
dfC2020$Maximum_Daily_Withdrawal <-
  as.numeric(gsub(",", "", dfC2020$Maximum_Daily_Withdrawal))
```

```
#combine with Durham data collected above in a plot
```

```
#first use rbind to combine the two dataframes
```

```
df_combined = rbind(df2020, dfC2020)
```

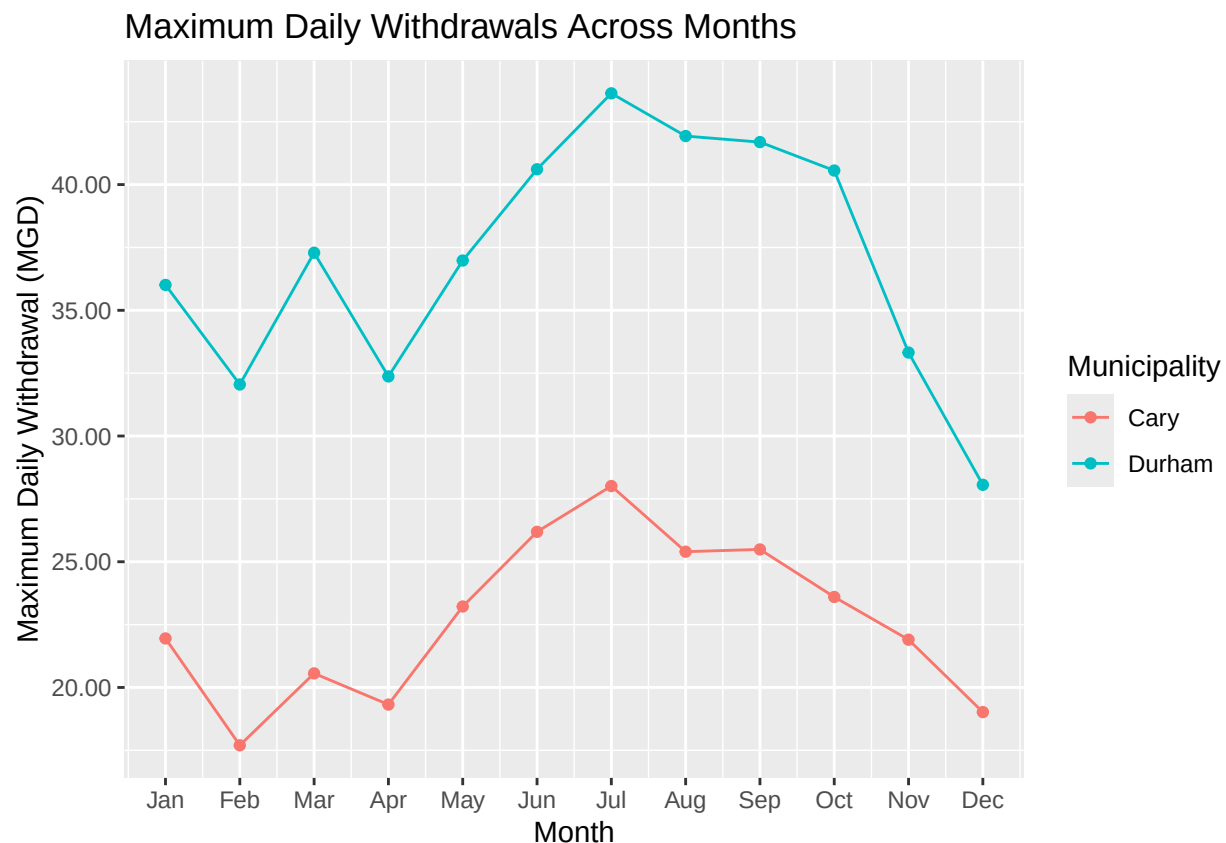
```
df_withdrawals_2020_2 <-
```

```
  ggplot(df_combined, aes(x = Month, y = Maximum_Daily_Withdrawal,
```

```

    group=Water_System_Name, color = Water_System_Name)) +
  geom_line() +
  geom_point() +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  scale_y_continuous(labels = scales::label_number(accuracy = 0.01)) + #fix this
  #making sure months show on x axis since I got points off for that last hw
  labs(
    title = "Maximum Daily Withdrawals Across Months",
    x = "Month",
    y = "Maximum Daily Withdrawal (MGD)",
    color = "Municipality")
show(df_withdrawals_2020_2)

```



- Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

TIP: See Section 3.2 in the "06_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs (where one of the two inputs does not change). Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

```

#9
#create a vector of all years we will use
the_years = c(2018, 2019, 2020, 2021, 2022, 2023)

```

```
#use map2 to iteratively run the function and combine dataframes using bind_rows
dfs_Cary <- map2(the_years, "03-92-020", scrape.it) %>%
  bind_rows()
```

```
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-92-020&year=2018"
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-92-020&year=2019"
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-92-020&year=2020"
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-92-020&year=2021"
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-92-020&year=2022"
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-92-020&year=2023"
```

```
show(dfs_Cary)
```

##	Month	Water_System_Name	Contact_Person	Ownership	Maximum_Daily-Withdrawal
## 1	1	Cary	Jamie Revels	Municipality	21.1900
## 2	5	Cary	Jamie Revels	Municipality	26.1300
## 3	9	Cary	Jamie Revels	Municipality	22.8800
## 4	2	Cary	Jamie Revels	Municipality	18.4800
## 5	6	Cary	Jamie Revels	Municipality	31.3300
## 6	10	Cary	Jamie Revels	Municipality	22.6500
## 7	3	Cary	Jamie Revels	Municipality	18.9300
## 8	7	Cary	Jamie Revels	Municipality	29.6900
## 9	11	Cary	Jamie Revels	Municipality	20.2000
## 10	4	Cary	Jamie Revels	Municipality	18.7700
## 11	8	Cary	Jamie Revels	Municipality	23.6500
## 12	12	Cary	Jamie Revels	Municipality	19.9900
## 13	1	Cary	Jamie Revels	Municipality	18.9900
## 14	5	Cary	Jamie Revels	Municipality	28.5900
## 15	9	Cary	Jamie Revels	Municipality	25.7600
## 16	2	Cary	Jamie Revels	Municipality	20.8500
## 17	6	Cary	Jamie Revels	Municipality	25.3100
## 18	10	Cary	Jamie Revels	Municipality	28.0400
## 19	3	Cary	Jamie Revels	Municipality	19.9600
## 20	7	Cary	Jamie Revels	Municipality	28.4200
## 21	11	Cary	Jamie Revels	Municipality	19.5900
## 22	4	Cary	Jamie Revels	Municipality	21.3400
## 23	8	Cary	Jamie Revels	Municipality	26.2800
## 24	12	Cary	Jamie Revels	Municipality	18.5100
## 25	1	Cary	Jamie Revels	Municipality	21.9500
## 26	5	Cary	Jamie Revels	Municipality	23.2200
## 27	9	Cary	Jamie Revels	Municipality	25.4900
## 28	2	Cary	Jamie Revels	Municipality	17.7000
## 29	6	Cary	Jamie Revels	Municipality	26.1900
## 30	10	Cary	Jamie Revels	Municipality	23.6000
## 31	3	Cary	Jamie Revels	Municipality	20.5600
## 32	7	Cary	Jamie Revels	Municipality	28.0100
## 33	11	Cary	Jamie Revels	Municipality	21.9000
## 34	4	Cary	Jamie Revels	Municipality	19.3200
## 35	8	Cary	Jamie Revels	Municipality	25.4000
## 36	12	Cary	Jamie Revels	Municipality	19.0200
## 37	1	Cary	Jamie Revels	Municipality	20.2500
## 38	5	Cary	Jamie Revels	Municipality	27.2100
## 39	9	Cary	Jamie Revels	Municipality	26.2600

## 40	2	Cary	Jamie Revels Municipality	23.1200
## 41	6	Cary	Jamie Revels Municipality	25.2800
## 42	10	Cary	Jamie Revels Municipality	24.9000
## 43	3	Cary	Jamie Revels Municipality	20.6100
## 44	7	Cary	Jamie Revels Municipality	25.9000
## 45	11	Cary	Jamie Revels Municipality	21.8200
## 46	4	Cary	Jamie Revels Municipality	23.2200
## 47	8	Cary	Jamie Revels Municipality	26.6500
## 48	12	Cary	Jamie Revels Municipality	23.4000
## 49	1	Cary	Jamie Revels Municipality	20.5900
## 50	5	Cary	Jamie Revels Municipality	23.2300
## 51	9	Cary	Jamie Revels Municipality	27.8800
## 52	2	Cary	Jamie Revels Municipality	20.0700
## 53	6	Cary	Jamie Revels Municipality	26.8100
## 54	10	Cary	Jamie Revels Municipality	25.7200
## 55	3	Cary	Jamie Revels Municipality	22.1800
## 56	7	Cary	Jamie Revels Municipality	29.0900
## 57	11	Cary	Jamie Revels Municipality	22.4300
## 58	4	Cary	Jamie Revels Municipality	22.3400
## 59	8	Cary	Jamie Revels Municipality	29.4400
## 60	12	Cary	Jamie Revels Municipality	20.2800
## 61	1	Cary	Jamie Revels Municipality	19.9200
## 62	5	Cary	Jamie Revels Municipality	25.3700
## 63	9	Cary	Jamie Revels Municipality	27.1600
## 64	2	Cary	Jamie Revels Municipality	19.7700
## 65	6	Cary	Jamie Revels Municipality	26.5500
## 66	10	Cary	Jamie Revels Municipality	26.3000
## 67	3	Cary	Jamie Revels Municipality	20.3800
## 68	7	Cary	Jamie Revels Municipality	28.1100
## 69	11	Cary	Jamie Revels Municipality	23.2000
## 70	4	Cary	Jamie Revels Municipality	21.2000
## 71	8	Cary	Jamie Revels Municipality	27.5800
## 72	12	Cary	Jamie Revels Municipality	19.5600
##	Year			
## 1	2018			
## 2	2018			
## 3	2018			
## 4	2018			
## 5	2018			
## 6	2018			
## 7	2018			
## 8	2018			
## 9	2018			
## 10	2018			
## 11	2018			
## 12	2018			
## 13	2019			
## 14	2019			
## 15	2019			
## 16	2019			
## 17	2019			
## 18	2019			
## 19	2019			
## 20	2019			

21 2019
22 2019
23 2019
24 2019
25 2020
26 2020
27 2020
28 2020
29 2020
30 2020
31 2020
32 2020
33 2020
34 2020
35 2020
36 2020
37 2021
38 2021
39 2021
40 2021
41 2021
42 2021
43 2021
44 2021
45 2021
46 2021
47 2021
48 2021
49 2022
50 2022
51 2022
52 2022
53 2022
54 2022
55 2022
56 2022
57 2022
58 2022
59 2022
60 2022
61 2023
62 2023
63 2023
64 2023
65 2023
66 2023
67 2023
68 2023
69 2023
70 2023
71 2023
72 2023

```

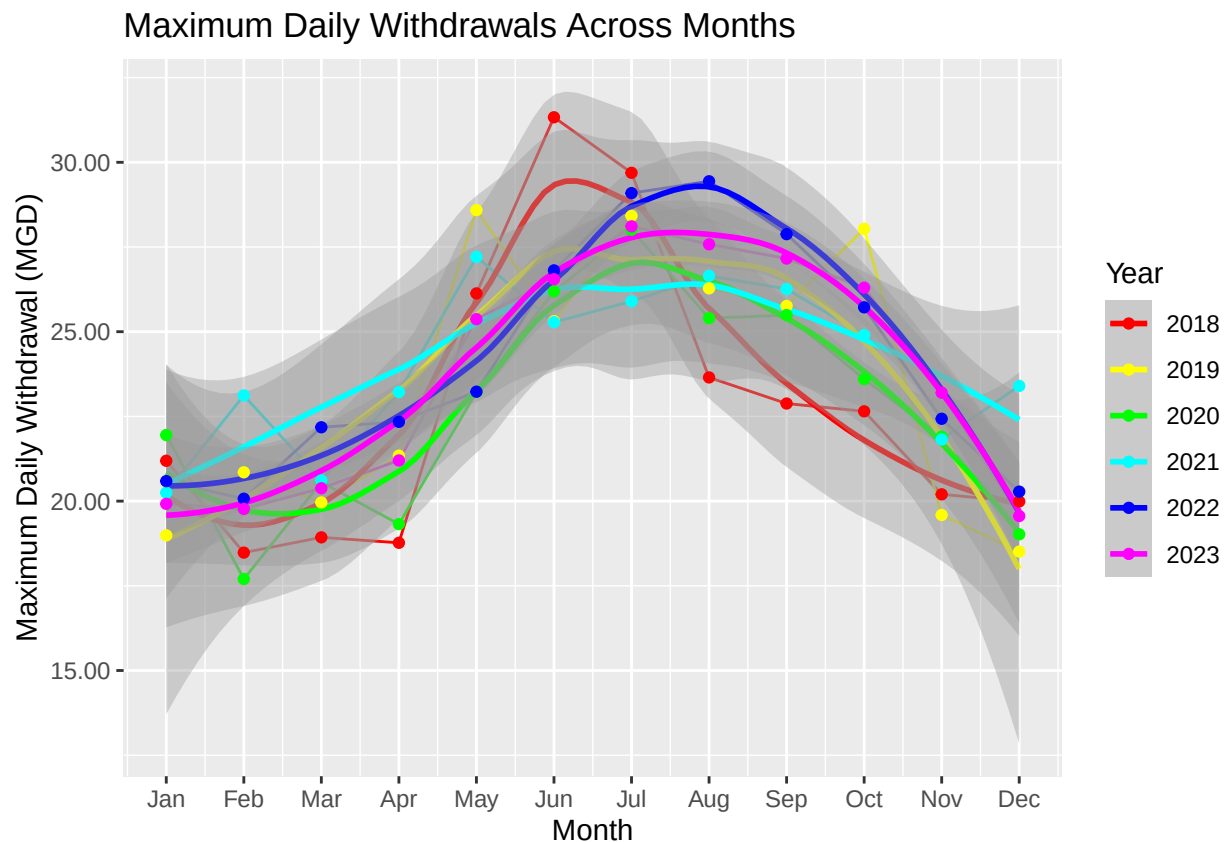
#fix maximum daily withdrawal column
dfs_Cary$Maximum_Daily-Withdrawal <-
  as.numeric(gsub(",", "", dfs_Cary$Maximum_Daily-Withdrawal))

#create plot, similar to same code above
dfC_plot <-
  ggplot(dfs_Cary, aes(x = Month, y = Maximum_Daily-Withdrawal, group=Year,
    color = factor(Year))) +

  geom_line() +
  geom_smooth(method = "loess") +
  geom_point() + #add rainbow color palette for clarity
  scale_color_manual(values = rainbow(length(unique(dfs_Cary$Year)))) +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  scale_y_continuous(labels = scales::label_number(accuracy = 0.01)) + #fix this
  #making sure months show on x axis since I got points off for that last hw
  labs(
    title = "Maximum Daily Withdrawals Across Months",
    x = "Month",
    y = "Maximum Daily Withdrawal (MGD)",
    color = "Year")
show(dfC_plot)

```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: Cary's water usage follows a similar year over year pattern. It is lowest from January through March, in roughly the 17.5-22.5 MGD range. By March it begins steadily increasing to a summer peak of 25-30 MGD, around June through August. At which point it begins a steady decrease again until December In December it reaches the same 17.5-22.5MGD range of values.